

Notes: Greenwood and Hanson (RFS, 2013)

Issuer Quality and Corporate Bond Returns

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1 Big picture

- **Question.** Does the *quality mix* of firms issuing corporate debt tell us when credit markets are overheated, and does it forecast subsequent corporate bond returns better than the *quantity* of aggregate credit?
- **Core idea.** Broad changes in the pricing of credit risk affect low-quality firms more than high-quality firms. When expected returns on credit are unusually low and credit is “cheap,” low-quality firms issue disproportionately more debt. So *who* is borrowing is an informative barometer of financing conditions.
- **Main empirical objects.** The paper builds two annual issuer-quality series:
 - ISS_t^{EDF} : a Compustat-based measure, 1962–2008, comparing the expected default frequency of high versus low net debt issuers;
 - HYS_t : the speculative-grade share of nonfinancial bond issuance, 1926–2008.
- **Main result.** When issuer quality is poor, risky corporate bonds subsequently underperform Treasuries. The forecasting power is strongest for high-yield bonds, weaker for BBB bonds, and weakest for AAA bonds.
- **Economic magnitude.** In the 1962–2008 sample, the univariate R^2 from forecasting cumulative high-yield excess returns with ISS^{EDF} reaches about 12% at a 1-year horizon, 26% at 2 years, and 29% at 3 years. A one-standard-deviation rise in ISS^{EDF} lowers next year’s high-yield excess return by about 4.5 percentage points.
- **Quality beats quantity.** Aggregate credit growth does forecast returns, but much of that forecasting power comes from debt growth among low-quality firms. Once the quality mix is included, aggregate quantity has much less incremental power.
- **Decomposition.** Poor issuer quality predicts both future *spread widening* and *higher defaults*. So the low subsequent returns to corporate bonds are not just a valuation quirk in secondary-market prices; they also reflect ex post deterioration in fundamentals.
- **Why the paper matters.** A lot of macro and corporate-finance work studies credit cycles through borrower net worth, collateral, or intermediary capital. This paper adds a different margin: time variation in investor risk appetite and beliefs. The composition of debt issuance contains information about the time-varying price of credit risk.
- **Interpretation.** The paper does not claim issuer quality *causes* low future returns. Rather, issuer quality is an equilibrium outcome that reflects financing conditions. Firms respond to cheap credit by issuing more, and lower-quality firms respond the most.
- **Policy takeaway.** Looking only at credit quantities or credit spreads is not enough. Policymakers and investors should also look at the credit quality of the firms receiving finance.

2 Data and measurement (key objects)

2.1 Main Compustat-based sample: 1962–2008

- The central issuer-quality series is built from annual Compustat firm data.

- The key firm-level financing variable is net debt issuance scaled by lagged assets:

$$d_{it} = \frac{D_{it}}{A_{i,t-1}}, \quad D_{it} \equiv \Delta \text{Assets}_{it} - \Delta \text{Book Equity}_{it}. \quad (1)$$

- High debt issuers are firms in the top NYSE quintile of d_{it} in year t .
- Low debt issuers are firms in the bottom NYSE quintile of d_{it} in year t .

Interpretation. The paper is not asking whether a given firm issues debt. It asks whether, in a given year, the *firms doing the most borrowing* are unusually risky relative to the firms doing the least borrowing.

2.2 Expected default frequency (EDF)

The paper proxies for firm credit quality using the Merton-style expected default frequency, computed following Bharath and Shumway (2008). Conceptually, EDF is higher for firms with:

- higher leverage,
- higher asset volatility,
- lower distance to default.

The authors use *EDF deciles* rather than raw EDF levels in the baseline construction.

Why deciles?

- to reduce the influence of outliers,
- to reduce sensitivity to secular changes in the cross-sectional dispersion of EDF,
- to make the units easy to interpret.

2.3 Primary issuer-quality measure: ISS^{EDF}

The baseline annual issuer-quality measure is

$$\text{ISS}_t^{\text{EDF}} = \overline{\text{EDF}}_t^{\text{High debt issuers}} - \overline{\text{EDF}}_t^{\text{Low debt issuers}}, \quad (2)$$

where the averages are taken across firms in the top and bottom NYSE quintiles of net debt issuance.

Interpretation.

- High values of $\text{ISS}_t^{\text{EDF}}$ mean that firms issuing the most debt are of especially poor credit quality.
- A value of 1 means high issuers are, on average, one EDF decile riskier than low issuers.
- This is a *quality-mix* variable: it captures the composition of issuance, not the total volume.

2.4 Alternative issuer-quality measures

The paper constructs other time-series measures of issuer quality using the same high-versus-low issuance logic but different firm characteristics:

- Shumway bankruptcy hazard,
- interest coverage,
- leverage,
- idiosyncratic volatility,
- firm size,
- firm age,
- dividend payer status.

Interpretation. If the main result is really about credit quality, it should not depend on one very specific proxy such as EDF. The paper shows that it does not.

2.5 Bond-market-based quality measure: the high-yield share

A second issuer-quality measure is built from ratings on new bond issues:

$$\text{HYS}_t = \frac{\sum_{i \in \text{HighYield}} B_{it}}{\sum_{i \in \text{HighYield}} B_{it} + \sum_{i \in \text{InvGrade}} B_{it}}, \quad (3)$$

where B_{it} is the principal amount of issue i in year t .

In forecasting regressions, the paper uses $\log(\text{HYS}_t)$.

Data construction.

- 1983–2008: FISD;
- 1966–1982: hand-collected Moody’s Bond Survey data;
- 1926–1965: historical NBER compilations from Hickman and Atkinson.

Why this measure is useful.

- It extends the analysis back to 1926.
- It is simple and directly tied to bond-market issuance quality.

Why the authors still prefer ISS^{EDF} .

- ISS^{EDF} covers loans as well as bonds, not just bond issuance.
- ISS^{EDF} is based on *net* debt issuance, not gross issuance.
- ISS^{EDF} keeps the quality concept more stable over time than ratings do.
- HYS can be noisy when aggregate bond issuance is very low, especially in 1933.

2.6 Returns

The key dependent variables are cumulative log excess returns on corporate bond indexes over Treasury benchmarks of similar maturity.

- High-yield excess returns:

$$rx_{t+1}^{HY} = r_{t+1}^{HY} - r_{t+1}^G.$$

- BBB excess returns:

$$rx_{t+1}^{BBB} = r_{t+1}^{BBB} - r_{t+1}^G.$$

- AAA excess returns:

$$rx_{t+1}^{AAA} = r_{t+1}^{AAA} - r_{t+1}^G.$$

- Multi-year cumulative returns are sums of annual log excess returns.

Sources.

- High-yield returns: Morningstar/Ibbotson through 1982; Barclays thereafter.
- BBB and AAA returns: Morningstar/Ibbotson for earlier decades; Barclays in later years.
- Treasury yields and term spreads: Ibbotson.

2.7 Other forecasting controls

The paper uses a rich control set:

- short Treasury yield $y_{S,t}^G$,
- term spread $y_{L,t}^G - y_{S,t}^G$,
- BBB credit spread $y_{L,t}^{BBB} - y_{L,t}^G$,
- lagged high-yield excess returns,
- industrial production growth,
- consumption growth,
- recession indicator,
- output gap,
- Lettau–Ludvigson cay,
- Ludvigson–Ng macro factors,
- intermediary balance-sheet variables from the Flow of Funds.

Interpretation. The paper is not just showing that issuer quality is correlated with the business cycle. It asks whether issuer quality has incremental forecasting content after controlling for the standard bond-market and macro-finance predictors.

3 Framework and empirical specifications (all equations + interpretation)

3.1 A simple model of credit-market pricing

Let $s_{\theta t}$ denote the promised spread paid by firm type θ in year t . The paper writes the spread as

$$s_{\theta t} = \pi_{\theta t} + \beta_{\theta} \delta_t, \quad (4)$$

where

- $\pi_{\theta t}$ is the objective probability of default,
- δ_t is the market-wide expected excess return on credit assets,
- β_{θ} captures how exposed a firm's financing cost is to shifts in the pricing of credit risk.

Expected excess returns satisfy

$$E_t[\text{rx}_{\theta, t+1}] = \beta_{\theta} \delta_t. \quad (5)$$

Interpretation.

- Credit spreads can be low for two very different reasons:
 - default probabilities are low;
 - required returns are low.
- The empirical problem is to separate these two forces.

3.2 Why low-quality firms are especially informative

Suppose there are low-risk firms L and high-risk firms H , with

$$\pi_{Lt} < \pi_{Ht} \quad \text{and} \quad \beta_L < \beta_H.$$

Thus, the financing costs of low-quality firms respond more strongly to shifts in market-wide credit pricing.

Each firm chooses debt issuance by trading off cheap financing against deviation from target leverage:

$$\max_{d_{\theta t}} \left\{ -\beta_{\theta} \delta_t d_{\theta t} - \frac{\gamma}{2} (d_{\theta t} - (\xi_t + \varepsilon_{\theta t}))^2 \right\}, \quad (6)$$

where

- ξ_t is a common target-leverage component,
- $\varepsilon_{\theta t}$ is a type-specific target-leverage component,
- γ is the cost of deviating from target leverage.

The optimal issuance decision is

$$d_{\theta t}^* = \xi_t + \varepsilon_{\theta t} - (\beta_{\theta}/\gamma) \delta_t. \quad (7)$$

Interpretation.

- Firms issue more when expected returns on credit are low.
- Because $\beta_H > \beta_L$, low-quality firms respond more to cheap credit.
- This is the main reason issuer *quality* is informative about future returns.

3.3 Quantity versus quality

Aggregating across high- and low-quality firms gives

$$d_{Ht}^* + d_{Lt}^* = (2\xi_t + \varepsilon_{Ht} + \varepsilon_{Lt}) - \frac{\beta_H + \beta_L}{\gamma} \delta_t, \quad (8)$$

while the quality mix is

$$d_{Ht}^* - d_{Lt}^* = (\varepsilon_{Ht} - \varepsilon_{Lt}) - \frac{\beta_H - \beta_L}{\gamma} \delta_t. \quad (9)$$

Interpretation.

- Aggregate quantity loads on the common component ξ_t , which can move for reasons unrelated to expected returns.
- The difference between high- and low-quality issuance removes that common component.
- So the quality mix is often a cleaner signal of time variation in expected returns than total credit growth.

3.4 Continuous issuer-quality proxy

In the model, a general continuous quality proxy is

$$\text{ISS}_t = E_t[\beta_i \mid \text{High } d_{it}^*] - E_t[\beta_i \mid \text{Low } d_{it}^*]. \quad (10)$$

Interpretation. This captures the difference in the average risk sensitivity of heavy issuers versus light issuers. ISS^{EDF} is the empirical version of this idea, with EDF standing in for latent credit quality.

3.5 Forecasting regressions

The baseline predictive regression is

$$\text{rx}_{t+k}^{\text{HY}} = a + bX_t + u_{t+k}, \quad (11)$$

where $k \in \{1, 2, 3\}$ and X_t is either ISS_t^{EDF} or $\log(\text{HYS}_t)$.

The multivariate forecasting regression is

$$\text{rx}_{t+k}^{\text{HY}} = a + bX_t + c \left(y_{L,t}^G - y_{S,t}^G \right) + d y_{S,t}^G + e \left(y_{L,t}^{\text{BBB}} - y_{L,t}^G \right) + f \text{rx}_t^{\text{HY}} + u_{t+k}. \quad (12)$$

Interpretation.

- Equation (11) asks whether issuer quality alone forecasts future corporate bond returns.
- Equation (12) asks whether issuer quality adds information beyond standard term-structure and credit-spread variables.

3.6 Out-of-sample predictive power

To check whether the results survive genuine forecasting, the paper computes

$$R_{OS}^2 = 1 - \frac{\sum_{t=s+1}^T (rx_t - \hat{rx}_t)^2}{\sum_{t=s+1}^T (rx_t - \bar{rx}_t)^2}, \quad (13)$$

where $\hat{rx}_t$ is the fitted return from the predictive regression estimated through time $t - 1$, and $\bar{rx}_t$ is the historical mean return estimated through time $t - 1$.

Interpretation. Positive out-of-sample R^2 means the issuer-quality variable beats the naïve forecast that excess returns are just equal to their historical mean.

3.7 Decomposing excess returns into spreads and defaults

The paper notes the approximate decomposition

$$rx_{t+1} \approx (1 - DEF_{t+1}) [s_t - Dur \cdot (s_{t+1} - s_t)] - DEF_{t+1} \cdot LGD_{t+1}, \quad (14)$$

where s_t is the initial credit spread, Dur is duration, and LGD is loss given default.

Interpretation. Poor future returns can come from:

- rising spreads on nondefaulted bonds,
- actual defaults and loss severity,
- or both.

The paper shows that issuer quality predicts both channels.

3.8 Intermediary-balance-sheet regressions

To see whether issuer quality is just proxying for intermediary capital, the paper estimates

$$ISS_t^{EDF} = a + bZ_t + c(y_{L,t}^G - y_{S,t}^G) + dy_{S,t}^G + e(y_{L,t}^{BBB} - y_{L,t}^G) + frx_t^{HY} + \varepsilon_t, \quad (15)$$

and

$$rx_{t+2}^{HY} = a + b_1 ISS_t^{EDF} + b_2 Z_t + c(y_{L,t}^G - y_{S,t}^G) + dy_{S,t}^G + e(y_{L,t}^{BBB} - y_{L,t}^G) + frx_t^{HY} + u_{t+2}, \quad (16)$$

where Z_t denotes insurer, broker-dealer, or bank balance-sheet variables.

Interpretation. If intermediary capital is the fundamental driver, then once Z_t is controlled for, the predictive power of ISS_t^{EDF} should weaken substantially.

3.9 Determinants of issuer quality

The paper studies the time-series determinants of issuer quality using

$$ISS_t^{EDF} = a + by_{S,t}^G + c(y_{L,t}^G - y_{S,t}^G) + drx_{t-1}^{HY} + eDEF_t^{HY} + u_t, \quad (17)$$

and in changes,

$$\Delta_k \text{ISS}_t^{EDF} = a + b\Delta_k y_{S,t}^G + c\Delta_k (y_{L,t}^G - y_{S,t}^G) + d\text{rx}_{t-k-1 \rightarrow t-1}^{\text{HY}} + e\Delta_k \text{DEF}_t^{\text{HY}} + \Delta_k u_t. \quad (18)$$

Interpretation.

- Low or falling rates and a flat curve are consistent with “reaching for yield.”
- Low recent defaults and high recent returns are consistent with extrapolative beliefs.

4 Identification and empirical strategy (what makes the estimates credible)

4.1 What identifies the key forecasting relation?

The paper exploits time-series variation in the *composition* of debt issuance. The identifying logic is:

- broad changes in the pricing of credit risk affect all firms, but affect low-quality firms more;
- common shocks to investment opportunities or leverage targets affect the aggregate quantity of debt issuance;
- comparing high- and low-quality issuers differences out part of that common component.

Interpretation. The signal is not “credit is growing fast.” The signal is “credit is growing especially fast among risky borrowers.”

4.2 Why quality can add to spreads

A low credit spread can reflect

- genuinely low expected defaults,
- low required returns,
- or some combination of the two.

Issuer-quality variables can add beyond spreads because corporate financing decisions aggregate information about the *price* of credit that is hard for the econometrician to observe directly.

Interpretation. Firms behave as if they understand when credit is cheap. Their issuance choices therefore reveal information about expected returns beyond what is visible in bond spreads alone.

4.3 Why the result is not a mechanical composition effect

A simple composition story would say that when more low-rated bonds are in the index, the index should have higher required returns because it has higher risk exposure. That logic predicts the opposite sign of the main result.

Why this matters. The paper does not merely show that the high-yield index becomes riskier when low-quality firms issue more. It shows that such periods are followed by *lower*, sometimes even statistically significantly negative, excess returns.

4.4 Why the empirical strategy is credible

- **Two independent measures.** The result appears using both ISS^{EDF} and $\log(\text{HYS})$.
- **Multiple horizons.** Predictability is present at 1-, 2-, and 3-year horizons.
- **Cross-rating pattern.** Effects are strongest for high-yield bonds and weaker for BBB and AAA, matching the theory that low-grade bonds load more on shifts in credit pricing.
- **Rich controls.** Results survive standard yield variables, macro controls, cay, and Ludvigson–Ng factors.
- **Robust variable construction.** The findings survive alternative credit-quality proxies, alternative weighting schemes, and nonfinancial-only samples.
- **Out-of-sample forecasting.** The paper shows nontrivial out-of-sample predictive power.

4.5 Main limitations

- ISS^{EDF} starts in 1962, so the long historical analysis relies on HYS.
- Holding-period returns are measured on bond indexes by rating class, not on exact issuance cohorts.
- Time-series evidence alone cannot fully distinguish rational discount-rate variation, intermediary frictions, and mispricing.
- HYS can be distorted when total issuance is abnormally low, especially in 1933.

Interpretation. The paper is strongest as a measurement paper about credit-market overheating and a return-prediction paper about expected bond returns. It is necessarily more tentative when attributing the underlying mechanism.

5 Main results and interpretation

5.1 Issuer quality is a central feature of the credit cycle

The baseline series ISS^{EDF} is high in classic credit booms:

- the late 1960s,
- the late 1980s junk-bond boom,
- 1997–1998,
- and 2005–2007.

It tends to be low in or around recessions, though not perfectly so.

Economic meaning. Credit booms are not just periods of rapid borrowing. They are periods in which the marginal borrowers become worse.

5.2 Poor issuer quality predicts low future corporate bond returns

This is the paper’s headline finding.

- For high-yield bonds, the coefficient on ISS^{EDF} is about -9.53 at a 1-year horizon, -15.25 at 2 years, and -17.30 at 3 years.
- The corresponding univariate R^2 values are about 12%, 26%, and 29%.
- A one-standard-deviation increase in ISS^{EDF} lowers next year’s high-yield excess return by about 4.5 percentage points.

Interpretation. When risky firms are able to issue a lot of debt relative to safe firms, subsequent returns to holding risky corporate debt are poor.

5.3 The cross-rating pattern fits the model

The predictive coefficients decline in magnitude as bond quality improves:

- largest for high-yield,
- smaller for BBB,
- smallest for AAA.

Interpretation. This is exactly what the paper’s framework predicts: lower-rated bonds are more exposed to market-wide shifts in the pricing of credit risk.

5.4 The result survives standard yield and macro controls

When the forecasting regressions add the short rate, term spread, BBB spread, and lagged high-yield returns, the coefficient on issuer quality remains economically large and statistically strong.

Illustration.

- In the full 1962–2008 sample, the 2-year high-yield forecasting coefficient on ISS^{EDF} remains around -13.9 in the controlled regression.
- In the post-1983 sample, the 2-year coefficient remains around -15.9 .

Interpretation. Issuer quality is not just a relabeling of spreads or the slope of the yield curve.

5.5 The high-yield share also works, especially in the modern sample

$\log(\text{HYS})$ forecasts returns too, with weaker full-sample results and stronger modern-sample results.

Interpretation.

- This supports the paper’s central hypothesis using a completely different data source.
- The weaker long-run fit reflects limitations of HYS, especially the 1933 outlier and secular shifts between bond and loan markets.

5.6 Quality is more informative than quantity

Aggregate debt growth does negatively forecast high-yield excess returns, but issuer quality performs better in head-to-head tests.

- The correlation between ISS^{EDF} and Compustat aggregate debt growth is about 0.45.
- The correlation with Flow of Funds credit growth is even stronger.
- Yet once ISS^{EDF} is included, aggregate quantity has much less incremental predictive power.

Interpretation. Booms in total credit matter, but what really matters is whether the boom is concentrated in low-quality borrowers.

5.7 Low-quality firms drive the quantity effect

When the authors split firms into EDF quintiles and examine debt growth separately:

- debt growth among the safest firms predicts returns only weakly;
- debt growth among the riskiest firms predicts returns strongly.

A one-standard-deviation rise in debt growth for the safest EDF quintile lowers two-year future high-yield returns by about 3.47 percentage points; for the riskiest quintile, the effect is about 7.09 percentage points.

Interpretation. The credit cycle is not symmetric across firms. The most informative part of a credit boom is the expansion of financing to the weakest borrowers.

5.8 The result is robust

The return-predictive relation survives:

- pre-1983 and post-1983 subsamples,
- dropping 2008–2009,
- extending the Compustat series back to 1952,
- controlling for output gap, industrial production, consumption growth, recession dummies, cay, and macro factors,
- alternative constructions using raw EDF, long-term debt issuance, asset weighting, debt-issue weighting, and nonfinancial-only samples,
- alternate credit-quality measures such as Shumway distress and interest coverage.

Interpretation. This is not a fragile result driven by one sample split or one variable definition.

5.9 Issuer quality predicts spread changes and defaults

Poor issuer quality forecasts:

- future widening in BBB spreads,

- higher future issuer-weighted default rates on high-yield bonds.

Interpretation. The return predictability is not just telling us that bonds are expensive relative to Treasuries today. It also predicts that future credit outcomes will be worse.

5.10 The paper's interpretation of mechanisms

1. Rationally time-varying price of risk. Some evidence points in this direction:

- issuer quality is cyclical,
- macro controls reduce the coefficient somewhat,
- lower-rated bonds are more predictable than higher-rated bonds.

But the paper argues that this story is incomplete because:

- issuer quality sometimes forecasts statistically significantly negative excess returns;
- the variable does not line up cleanly with equity-premium forecasting or a fully integrated risk-premium story;
- out-of-sample predictive power is unusually large.

2. Intermediary balance sheets. Some balance-sheet proxies move in the right direction:

- poor issuer quality is associated with stronger insurer balance sheets,
- lagged bank stock returns and bank loan losses are correlated with issuer quality in the expected direction.

But controlling for these variables barely changes the predictive coefficient on ISS^{EDF} .

Interpretation. Intermediary capital may matter, but it does not subsume the issuer-quality signal.

3. Reaching for yield and overextrapolation. The paper finds that issuer quality is high when:

- short rates are low or falling,
- the term spread is low or flattening,
- past high-yield returns have been strong,
- past default rates have been low.

These patterns are especially strong in changes: past defaults and returns explain a large fraction of two-year changes in issuer quality.

Interpretation. Investors may become too optimistic after periods of benign credit experience, leading them to underprice risk and lend too easily to weak firms.

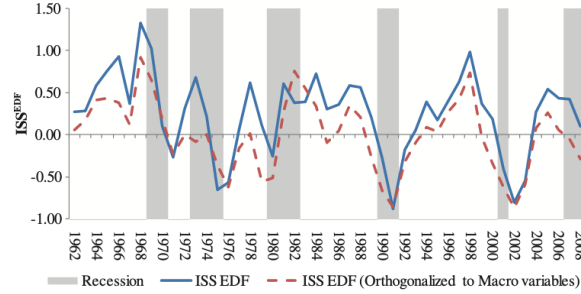


Figure 1
Issuer quality
 ISS^{EDF} is the difference between the average EDF decile between high- and low-debt issuers. EDF is the expected default frequency of Merton (1974). The figure also shows shading for NBER-designated recessions. The dotted line shows a version of ISS^{EDF} that has been orthogonalized with respect to the output gap (Hodrick-Prescott filtered real GDP).

6 Tables and figures

6.1 Figure 1: ISS^{EDF} over time

Read: Figure 1 plots the Compustat-based issuer-quality series from 1962 to 2008, with recessions shaded and an orthogonalized version that removes the output-gap component.

- Peaks occur in the late 1960s, late 1980s, 1997–1998, and 2005–2007.
- Troughs occur around 1970–1971, 1975–1976, the early 1990s, 2001–2002, and 2008.
- The orthogonalized series still looks very similar to the raw series.

Why it matters. The credit cycle is related to, but not identical to, the business cycle.

6.2 Table 1: Summary statistics

Read: Table 1 gives summary statistics for firm-level variables, issuer-quality time series, return series, macro controls, and intermediary-balance-sheet variables.

- Mean ISS^{EDF} is about 0.25 with standard deviation about 0.48.
- Mean spliced HYS is about 0.18.
- One-year high-yield excess returns are very volatile, with standard deviation above 13%.

Why it matters. The table shows both that issuer quality varies meaningfully over time and that there is enough return variation for economically large forecasting relations to matter.

6.3 Figure 2: The high-yield share

Read: Figure 2 plots the speculative-grade share of nonfinancial bond issuance from 1926 to 2008 alongside ISS^{EDF} .

- The series move together broadly.

Table 1
Summary statistics

	Mean	Median	SD	Min	Max
Panel A: Firm-level data (1962–2008, firm-year observations)					
Debt issues: d/A	0.09	0.04	0.40	−4.13	6.50
Equity issues: e/A	0.24	0.01	1.54	−0.55	48.09
External finance: $(d+e)/A$	0.30	0.09	1.31	−0.75	40.96
Expected default frequency: EDF	0.06	0.00	0.18	0.00	1.00
Shumway distress: $SHUM$	0.02	0.00	0.07	0.00	0.97
Interest coverage: $i/EBITDA$	23.72	5.44	223.97	−3,976.00	5,779.57
Leverage: D/A	0.23	0.20	0.19	0.00	0.85
Market capitalization: $Log(MV)$	1,166.39	75.62	8,367.79	0.11	602,432.90
Age	13.25	8.42	14.17	0.00	83.00
Dividends	0.02	0.00	0.03	0.00	0.60
Panel B: Debt issuer quality (1962–2008, annual time series)					
ISS^{EDF} Expected default frequency (high-low)	0.25	0.31	0.48	−0.88	1.32
ISS^{Shum} Shumway distress (high-low)	0.11	0.14	0.32	−0.79	0.78
ISS^{Incov} Interest coverage (low-high)	−0.22	−0.24	0.39	−0.98	0.81
ISS^{Lev} Leverage (low-high)	1.23	1.24	0.57	−0.03	2.58
ISS^{σ} CAPM σ (low-high)	−0.35	−0.30	0.54	−1.66	0.89
ISS^{ME} Size (small-large)	−0.60	−0.60	0.29	−1.24	0.04
ISS^{Age} Age (young-old)	0.59	0.58	0.24	−0.09	1.06
$ISS^{Nonpayer}$ Dividends (nonpayer-payer)	−0.06	−0.05	0.06	−0.20	0.08
Panel C: The high-yield share (1926–2008, annual time series)					
HYS (FISD, 1983–2008)	0.32	0.34	0.12	0.05	0.56
HYS (Moody's manuals and NBER, 1926–1982)	0.11	0.08	0.10	0.00	0.64
HYS (Spliced series, 1926–2008)	0.18	0.14	0.15	0.00	0.64
Panel D: Returns and macroeconomic controls % (1962–2008, annual time series)					
Macroeconomic controls:					
$y_{S,t}^G$	5.61	5.23	3.12	0.03	16.86
$y_{L,t}^G - y_{S,t}^G$	0.37	0.37	0.84	−1.73	2.28
$y_{t,t}^{BBB} - y_{t,t}^G$	1.69	1.60	0.77	0.50	5.11
Industrial production	2.76	3.06	4.38	−9.74	9.52
Aggregate consumption growth	3.36	3.46	1.81	−1.21	6.13
Recession dummy	0.29	0.00	0.46	0.00	1.00
Output gap (HP filtered log GDP)	0.02	0.08	1.37	−3.68	3.09
Cay	−0.01	0.01	1.52	−3.52	3.28
F_1 (1964–2007)	−6.21	−21.15	120.66	−198.90	490.20
F_2 (1964–2007)	12.55	23.84	113.24	−361.34	268.16
F_3 (1964–2007)	−12.57	−17.88	91.49	−373.68	250.01
Returns:					
r_{t+1}^{HY} (1-year excess high-yield return)	0.45	1.98	13.10	−42.65	48.31
r_{t+1}^{BBB} (1-year excess BBB return)	0.53	0.64	7.02	−22.42	27.91
r_{t+1}^{AAA} (1-year excess AAA return)	−0.51	0.39	3.40	−8.44	4.22
r_{t+2}^{HY} (2-year excess high-yield return)	−0.23	1.42	14.34	−50.37	31.38
r_{t+2}^{BBB} (2-year excess BBB return)	0.42	2.61	7.93	−27.52	20.27
r_{t+2}^{AAA} (2-year excess AAA return)	−1.05	0.59	5.65	−13.59	7.91
r_{t+3}^{HY} (3-year excess high-yield return)	−0.13	−0.52	15.80	−42.28	36.42
r_{t+3}^{BBB} (3-year excess BBB return)	0.79	2.22	8.98	−26.35	25.56
r_{t+3}^{AAA} (3-year excess AAA return)	−1.46	0.16	7.33	−20.30	9.98

(continued)

Table 1
Continued

	Mean	Median	SD	Min	Max
Panel D: Returns and macroeconomic controls % (1962–2008, annual time series)					
Returns:					
$MKTRF_{t+1}$	3.46	8.93	17.65	−51.59	27.41
HML_{t+1}	4.65	6.73	12.23	−30.58	32.95
SMB_{t+1}	2.40	3.69	12.37	−29.02	34.88
r_{t+1}^{EDF}	−1.89	8.93	13.14	−32.93	30.12
Default rates and changes in spreads:					
DEF_{t+1}^{HY}	3.36	2.09	3.10	0.00	12.97
$\Delta_1(y_{L,t+k}^{BBB} - y_{L,t+k}^G)$	0.02	−0.05	0.88	−3.41	3.07
$\Delta_2(y_{L,t+k}^{BBB} - y_{L,t+k}^G)$	0.11	0.06	0.95	−1.41	3.87
$\Delta_3(y_{L,t+k}^{BBB} - y_{L,t+k}^G)$	0.15	0.19	1.01	−1.89	3.49
Panel E: Intermediary balance sheets % (1962–2008, annual time series)					
$E/ABank$	5.83	5.21	2.88	1.70	12.10
$dA/ABank$	8.21	8.57	3.68	−1.39	15.26
$E/AINsurer$	12.20	12.30	1.35	8.58	14.27
$dA/AINsurer$	8.11	8.29	4.09	−8.00	15.46
E/ABD	6.98	4.93	4.77	0.47	18.14
dA/ABD	14.02	13.58	19.40	−28.29	74.11
$R_{Bank,t-3,t}$	30.79	36.29	40.36	−99.42	124.55
$LoanLosses_{Bank}$	0.75	0.58	0.64	0.12	3.53

Mean, median, standard deviation, and extreme values of firm characteristics (Panel A) and time-series variables (Panels B–E). Panel A summarizes the firm-level characteristics, with the exact details of their construction given in the Internet Appendix. Net debt issuance is the change in assets minus the change in book equity, scaled by lagged assets. Net equity issuance is the growth of balance sheet equity, net of retained earnings, scaled by lagged assets. External finance is the sum of net debt and net equity issuance. EDF is the Merton (1974) expected default frequency, calculated following Bharath and Shumway (2008). $SHUM$ is the bankruptcy hazard rate as estimated by Shumway (2001). Interest coverage is $EBITDA$ divided by annual interest expense. Leverage is book debt over assets. Age is the number of years on CRSP. Dividends is annual cash dividend scaled by assets. For each characteristic X , ISS^X compares the average characteristic decile of high and low quintile debt issuers in that year, where quintiles are based on NYSE breakpoints. These time-series are summarized in Panel B. Characteristics include EDF , $SHUM$, interest coverage, leverage, idiosyncratic volatility, size, age, and dividend policy. “High” and “Low” are defined such that high is associated with a higher default probabilities. Panel C summarizes the high yield share HYS , which is the dollar fraction of non-financial debt issues receive a high yield rating, according to Moody’s. From 1983–2008, HYS is based on data from the Fixed Income Securities Database (FISD). HYS from 1966–1982 is based on our assembly of bond issues from the Moody’s Bond Surveys, HYS from 1944–1965 is based on Table B-1 of Atkinson (1967), and HYS from 1926–1943 is based on Table V2 of Hickman (1960). Panel D summarizes bond returns and the time-series control variables. y_S the log yield on the short-term government bond, $y_{GL} - y_{GS}$ is the spread between the yields on the intermediate- and short-term government bonds from Ibbotson, and $y_{BBB} - y_{GL}$ is the BBB credit spread. Growth in industrial production, aggregate consumption growth, a recession dummy, the output gap, and Lettau and Ludvigson’s (2001) consumption wealth ratio (cay) are macroeconomic control variables. Log excess returns on corporate bonds are computed for bond index returns based on high yield (HY), BBB-rated, and AAA-rated bonds, and are denoted rx . Excess stock returns ($MKTRF$), HML returns, and SMB are from Ken French. Panel E summarizes time-series measures of intermediary balance sheet strength, including bank balance sheet capital (total assets minus liabilities all over assets) and bank balance sheet growth (percentage change in total assets); insurer balance sheet capital (total assets minus liabilities all over assets) and insurer balance sheet growth (percentage change in total assets); broker dealer balance sheet capital (total assets minus liabilities all over assets) and balance sheet growth (percentage change in total assets); lagged equal weighted returns on bank stocks based on Fama and French 48-industry classifications; and bank loan loss provisions scaled by total loans and leases from Table CB11 from the FDIC’s Historical Statistics on Banking. Bank, Insurer, and Broker Dealer data are from the Flow of Funds accounts. Banks include commercial banks (Table L109), saving institutions (L114) and credit unions (L115). Insurance companies include property and casualty (Table L116) and life (Table L117). Broker dealers are on Table L129.

- There is a clear regime shift after the early 1980s: the level and volatility of HYS increase sharply.
- The correlation between HYS and ISS^{EDF} is moderate before 1982 and stronger after 1982.

Why it matters. The historical bond-market measure and the Compustat-based measure tell a similar story, but the comparison also shows why the paper prefers ISS^{EDF} as the cleaner benchmark measure.

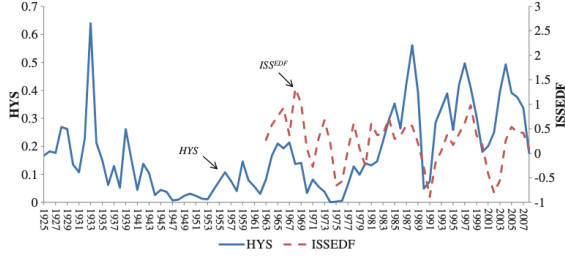


Figure 2
The high-yield share
HYS is the log fraction of nonfinancial corporate bond issuance with a high-yield rating from Moody's. HYS is constructed using data from Hickman (1960) and Atkinson (1967) NBER studies from 1926–1965, from hand-collected data from Moody's Bond Surveys from 1966–1982, and from FISD for 1983–2008. For comparison, the figure plots ISS^{EDF} on the right-hand scale.

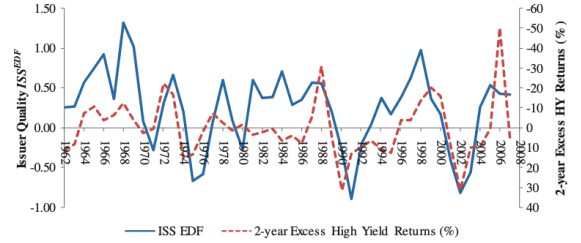


Figure 3
Issuer quality and subsequent high-yield excess returns
Issuer quality (left axis) plotted alongside cumulative excess high-yield bond returns for the following two years (right axis). Returns are plotted in reverse scale, so the negative correlation appears positive visually. Issuer quality is measured with ISS^{EDF} , the difference between the average EDF decile of high- and low-debt issuers from 1962–2008.

6.4 Figure 3: Issuer quality and future high-yield returns

Read: Figure 3 plots ISS^{EDF} against cumulative high-yield excess returns over the following two years, with returns on a reverse scale.

- The correlation is about -0.51 .
- High issuer-quality deterioration lines up with poor subsequent credit returns.

Why it matters. This is the visual headline result of the paper.

6.5 Table 3: Univariate return predictability

Read: Table 3 runs univariate forecasting regressions for high-yield, BBB, and AAA returns using ISS^{EDF} and $\log(HYS)$.

- High-yield: coefficient on ISS^{EDF} is about -9.53 , -15.25 , and -17.30 at 1-, 2-, and 3-year horizons.
- Corresponding R^2 values are about 12%, 26%, and 29%.
- The predictive coefficients shrink moving from high-yield to BBB to AAA.

Why it matters. The table establishes that the result is strong, persistent, and concentrated where the model says it should be: in lower-grade credit.

6.6 Table 4: Multivariate forecasting regressions

Read: Table 4 adds the short rate, term spread, BBB spread, and lagged high-yield returns.

Table 3
Issuer quality and the returns to corporate credit

	$X_t = ISS^{EDF}$						$X_t = \log(HYS)$					
	Full sample (1962–2008)			1983+			Full sample (1926–2008)			1983+		
	1-yr.	2-yr.	3-yr.	1-yr.	2-yr.	3-yr.	1-yr.	2-yr.	3-yr.	1-yr.	2-yr.	3-yr.
Panel A: High-yield excess returns (ex^{HY})												
b	-9.534	-15.254	-17.301	-11.152	-21.021	-27.966	-1.517	-2.917	-3.884	-11.483	-14.264	-17.798
[b]	[-3.971]	[-5.291]	[-5.688]	[-2.661]	[-5.201]	[-6.191]	[-1.771]	[-1.981]	[-1.891]	[-2.771]	[-4.231]	[-5.761]
R^2	0.12	0.26	0.29	0.10	0.32	0.48	0.02	0.04	0.05	0.15	0.21	0.28
Panel B: BBB excess returns (ex^{BBB})												
b	-5.311	-6.945	-6.645	-3.576	-6.002	-8.022	-0.874	-1.656	-2.100	-3.139	-3.344	-5.118
[b]	[-3.586]	[-4.471]	[-3.008]	[-1.871]	[-3.011]	[-3.261]	[-1.711]	[-1.861]	[-1.851]	[-1.332]	[-1.246]	[-2.091]
R^2	0.13	0.18	0.13	0.04	0.15	0.23	0.02	0.04	0.05	0.05	0.06	0.13
Panel C: AAA excess returns (ex^{AAA})												
b	-2.278	-3.321	-3.372	-0.799	-2.144	-3.578	-0.310	-0.473	-0.108	-0.030	-1.015	-1.020
[b]	[-2.401]	[-2.251]	[-1.201]	[-0.931]	[-1.381]	[-2.031]	[-1.171]	[-1.031]	[-0.191]	[-0.071]	[-1.1071]	[-1.141]
R^2	0.10	0.08	0.05	0.02	0.07	0.13	0.02	0.01	0.00	0.00	0.02	0.05

Univariate time-series forecasting regressions of log excess returns on issuer quality ISS^{EDF} and $\log(HYS)$:

$$r_{X_t+k} = a + b \cdot X_t + u_{t+k}$$

ISS^{EDF} is the difference between the expected default frequency deciles of high and low net debt issuers in each year. The high yield share (HYS) is the fraction of nonfinancial corporate bond issuance with a high-yield rating from Moody's. In Panel A, the dependent variable is the cumulative 1-, 2-, or 3-year excess return on high-yield bonds. In Panel B, the dependent variable is the cumulative 1-, 2-, or 3-year excess return on BBB-rated corporate bonds. In Panel C, the dependent variable is the cumulative 1-, 2-, or 3-year excess return on AAA-rated corporate bonds. t -statistics for k -period forecasting regressions (in brackets) are based on Newey-West (1987) standard errors, allowing for serial correlation up to k -lags.

Table 4
Multivariate forecasting regressions

	1-yr. returns			2-yr. returns			3-yr. returns		
Panel A: $X_t = ISS^{EDF}$ (1962–2008)									
ISS^{EDF}	-7.636 [−3.45]	-8.617 [−2.97]	-6.282 [−2.40]	-11.022 [−3.45]	-18.052 [−4.60]	-13.890 [−4.54]	-14.214 [−2.57]	-21.697 [−3.83]	-19.343 [−3.77]
$y_{L,t}^G - y_{S,t}^G$	1.495 [0.62]	2.031 [0.67]	5.025 [2.25]		8.055 [3.36]	4.61 [2.49]		9.477 [5.68]	
$y_{S,t}^G$	-0.442 [−0.62]	-0.49 [−0.68]	0.487 [1.02]		0.845 [1.40]	1.152 [1.86]		2.102 [3.06]	
$y_{L,t}^{BBB} - y_{L,t}^G$		3.836 [1.29]	3.773 [1.21]		-1.498 [−0.52]	-5.05 [−1.59]		-3.595 [0.86]	-10.624 [−2.44]
rx_t^{HY}		-0.264 [−1.77]	-0.29 [−1.62]		-0.498 [−2.35]	-0.729 [−3.46]		-0.667 [3.45]	-0.936 [−5.20]
R^2	0.15	0.28	0.32	0.31	0.34	0.45	0.33	0.39	0.51
Panel B: $X_t = ISS^{EDF}$ (1983–2008)									
ISS^{EDF}	-6.120 [−1.07]	-7.609 [−1.50]	-4.492 [−0.85]	-15.578 [−2.67]	-24.240 [−5.33]	-15.865 [−2.49]	-31.055 [−4.60]	-35.796 [−5.78]	-35.669 [−7.09]
$y_{L,t}^G - y_{S,t}^G$	2.537 [0.52]	4.478 [0.89]	6.687 [1.57]		10.175 [2.16]	3.773 [1.41]		7.136 [3.08]	
$y_{S,t}^G$	-1.036 [−0.60]	0.392 [0.27]	0.608 [0.67]		0.262 [0.26]	2.874 [3.02]		2.566 [2.90]	
$y_{L,t}^{BBB} - y_{L,t}^G$		9.948 [2.00]	9.214 [1.88]		-4.242 [−0.48]	-6.044 [−0.70]		-15.979 [−1.83]	-17.903 [−2.11]
rx_t^{HY}		-0.020 [−0.08]	-0.114 [−0.45]		-0.536 [−2.29]	-0.773 [−3.07]		-0.823 [−4.80]	-0.899 [−7.66]
R^2	0.13	0.33	0.35	0.37	0.39	0.51	0.56	0.60	0.70
Panel C: $X_t = \log(HYS)$ (1926–2008)									
$\log(HYS)$	-1.495 [−1.88]	-1.925 [−2.32]	-1.717 [−2.14]	-3.154 [−2.51]	-3.438 [−2.57]	-3.250 [−2.63]	-4.067 [−2.21]	-4.543 [−2.23]	-4.254 [−2.20]
$y_{L,t}^G - y_{S,t}^G$	2.201 [1.20]	0.275 [0.12]	6.269 [2.47]		5.577 [1.46]	5.555 [1.31]		3.076 [0.51]	
$y_{S,t}^G$	-0.375 [−0.78]	-0.514 [−0.94]	-0.054 [−0.07]		-0.057 [−0.06]	-0.240 [−0.20]		-0.515 [−0.38]	
$y_{L,t}^{BBB} - y_{L,t}^G$		3.492 [1.53]	2.936 [1.13]		4.552 [1.60]	1.024 [0.28]		5.904 [1.78]	3.658 [0.76]
rx_t^{HY}		0.172 [1.06]	0.131 [0.80]		0.307 [1.27]	0.130 [0.49]		0.201 [0.73]	0.067 [0.21]
R^2	0.07	0.07	0.09	0.14	0.09	0.14	0.11	0.10	0.12
Panel D: $X_t = \log(HYS)$ (1983–2008)									
$\log(HYS)$	-11.180 [−2.89]	-6.632 [−1.58]	-5.326 [−1.20]	-11.889 [−3.21]	-12.432 [−3.12]	-9.525 [−2.32]	-15.000 [−5.05]	-17.516 [−4.02]	-14.451 [−5.19]
$y_{L,t}^G - y_{S,t}^G$	1.803 [0.39]	4.203 [0.81]	8.188 [2.10]		11.651 [2.55]	8.269 [2.34]		11.254 [3.52]	
$y_{S,t}^G$	-1.751 [−1.19]	-0.225 [−0.15]	-1.052 [−1.14]		-1.173 [−1.11]	-0.105 [−0.08]		-0.430 [−0.34]	
$y_{L,t}^{BBB} - y_{L,t}^G$		11.227 [2.42]	8.335 [1.56]		5.693 [0.53]	-4.489 [−0.57]		-0.161 [−0.01]	-9.068 [−0.85]
rx_t^{HY}		0.138 [0.56]	-0.059 [−0.27]		-0.025 [−0.08]	-0.620 [−2.20]		-0.077 [−0.22]	-0.585 [−2.79]
R^2	0.24	0.33	0.36	0.40	0.22	0.48	0.39	0.28	0.45

Time-series forecasting regressions of log excess returns on speculative-grade bonds on measures of debt issuance quality, controlling for the term spread, short-rate, credit spread, and lagged excess returns:

$$r_{X_t+k}^{HY} = a + b \cdot X_t + c \cdot (y_{L,t}^G - y_{S,t}^G) + d \cdot y_{S,t}^G + e \cdot (y_{L,t}^{BBB} - y_{L,t}^G) + f \cdot rx_t^{HY} + u_{t+k}$$

In Panel A, X_t is ISS^{EDF} from 1962–2008; in Panel B, X_t is ISS^{EDF} from 1983–2008; in Panel C, X_t is $\log(HYS)$ from 1926–2008; and in Panel D, X_t is $\log(HYS)$ from 1983–2008. t -statistics for k -period forecasting regressions (in brackets) are based on Newey-West (1987) standard errors, allowing for serial correlation up to k -lags.

- ISS^{EDF} remains strongly negative.
- The incremental explanatory power remains sizable.
- For high-yield returns, the R^2 rises substantially once controls are added.

Why it matters. Issuer quality is not redundant with standard forecasting variables from bond markets.

6.7 Figure 4 and Table 5: Quantity versus quality

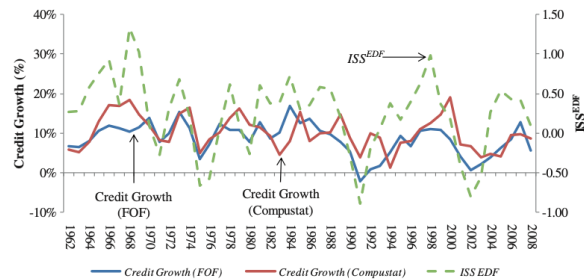


Figure 4
Quality and quantity and the credit cycle
The figure plots the Flow of Funds (FOF) and Compustat-based measures of aggregate nonfinancial credit growth on the left axis versus our measures of issuer quality, ISS^{EDF} , on the right axis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: Univariate											
<i>Issuer quality</i>	$R^{2\text{EDF}}$	-15.254 [-5.29]		-12.978 [-3.78]							
<i>Agg. debt growth</i>	$\Delta D_{\text{agg}}/D_{\text{agg}}$		-5.212 [-3.97]	-1.481 [-1.49]							
<i>Low EDF</i>	$\Delta D_1/D_1$			-3.474 [-2.04]					-1.565 [-0.86]	-4.917 [-3.16]	
2	$\Delta D_2/D_2$				-3.806 [-2.97]						
3	$\Delta D_3/D_3$					-2.842 [-2.31]					
4	$\Delta D_4/D_4$						-4.025 [-1.92]				
<i>High EDF</i>	$\Delta D_5/D_5$							-7.791 [-2.06]	-6.631 [-3.09]		
<i>High-low</i>	$\Delta D_5/D_5 - \Delta D_1/D_1$								-5.420 [-2.29]	-6.538 [-3.09]	
	R^2	0.26	0.13	0.29	0.06	0.07	0.04	0.08	0.24	0.26	0.14
											0.26
Panel B: Including additional time-series controls											
<i>Issuer quality</i>	$R^{2\text{EDF}}$	-13.800 [-4.54]		-11.442 [-3.73]							
<i>Agg. debt growth</i>	$\Delta D_{\text{agg}}/D_{\text{agg}}$		-6.497 [-3.22]	-5.126 [-2.40]							
<i>Low EDF</i>	$\Delta D_1/D_1$				-3.329 [-1.77]				-2.157 [-1.23]	-5.808 [-2.49]	
2	$\Delta D_2/D_2$					-2.367 [-1.17]					
3	$\Delta D_3/D_3$						-3.104 [-1.96]				
4	$\Delta D_4/D_4$							-4.629 [-2.68]			
<i>High EDF</i>	$\Delta D_5/D_5$								-7.053 [-2.31]	-6.827 [-3.04]	
<i>High-low</i>	$\Delta D_5/D_5 - \Delta D_1/D_1$									-5.429 [-2.60]	-6.731 [-3.04]
	R^2	0.45	0.41	0.51	0.34	0.33	0.34	0.39	0.51	0.52	0.44

Annual time-series regressions of the form $r_{i,t+1}^{\text{EDF}} = a + b \cdot X_{i,t} + c \cdot (V_{i,t}^{\text{EDF}} - V_{i,t-1}^{\text{EDF}}) + d \cdot (R_{i,t}^{\text{agg}} - R_{i,t-1}^{\text{agg}}) + e \cdot (R_{i,t}^{\text{Low}} - R_{i,t-1}^{\text{Low}}) + f \cdot (R_{i,t}^{\text{High}} - R_{i,t-1}^{\text{High}}) + \text{error}$, where $R_{i,t}^{\text{EDF}}$ is the cumulative two-tenor excess return on high-yield bonds, $\Delta D_{\text{agg}}/D_{\text{agg}}$ is the annual percentage change in total debt for nonfinancial Corporate firms, and $\Delta D_1/D_1$, $\Delta D_2/D_2$, $\Delta D_3/D_3$, $\Delta D_4/D_4$, $\Delta D_5/D_5$ denote aggregate debt growth in NYSE:EDF deciles 4 based on $R_{i,t}^{\text{EDF}}$ values. Panel B shows regressions with controls: the term spread, short rate, credit spread, and lagged excess high-yield returns. t -statistics (in brackets) are based on Newey-West (1994) standard errors, allowing for serial correlation up to two lags.

Read: Figure 4 shows that issuer quality and aggregate credit growth move together. Table 5 then asks which one actually forecasts returns better.

- Aggregate credit growth is informative.
- But issuer quality wins the horserace.
- Debt growth among low-quality firms is much more predictive than debt growth among high-quality firms.

Why it matters. Credit booms become especially dangerous when the extra borrowing is concentrated in weak issuers.

6.8 Table 6: Robustness

Read: Table 6 reruns the main 2-year high-yield forecasting regression across many specifications.

- The main coefficient remains negative and significant across subsamples.
- It survives controls for business-cycle variables, cay, macro factors, and aggregate leverage.
- Alternative issuer-quality constructions also work.

Why it matters. The result is not an artifact of one particular empirical design choice.

Table 6
Robustness

		Univariate			With time-series controls		
		<i>b</i>	[<i>t</i>]	R^2	<i>b</i>	[<i>t</i>]	R^2
(1)	Baseline result	-15.254	[-5.29]	0.26	-13.890	[-4.54]	0.45
Subsamples							
(2)	Longer 1952–2008 sample	-9.403	[-3.03]	0.16	-8.367	[-3.00]	0.38
(3)	Pre-1983	-8.994	[-3.90]	0.23	-9.383	[-2.48]	0.40
(4)	1983–present	-21.021	[-5.20]	0.32	-15.865	[-2.49]	0.51
(5)	Dropping 2008–2009	-14.556	[-5.32]	0.33	-13.997	[-4.33]	0.47
Additional controls							
(6)	Output gap	-13.592	[-4.05]	0.27	-13.548	[-4.32]	0.46
(7)	Other macro variables	-12.975	[-2.97]	0.30	-10.045	[-3.08]	0.50
(8)	Leads/lags of macro variables	-11.834	[-2.67]	0.46	-11.639	[-2.36]	0.59
(9)	Eight macrofactors	-12.303	[-4.29]	0.48	-13.659	[-4.13]	0.56
(10)	<i>Cay</i>	-12.910	[-3.99]	0.41	-11.550	[-3.11]	0.49
(11)	Cochrane-Piazzesi tent factor	-14.499	[-4.87]	0.30	-13.818	[-3.66]	0.45
(12)	Aggregate corporate leverage	-15.252	[-5.21]	0.26	-14.006	[-4.25]	0.46
Alternate constructions of ISS^{EDF}							
(13)	Debt Issues, <i>EDF</i> level	-372.414	[-5.37]	0.26	-274.561	[-6.72]	0.44
(14)	Long-term debt Issues, <i>EDF</i> decile	-18.663	[-4.28]	0.24	-15.087	[-4.21]	0.44
(15)	Short-term debt Issues, <i>EDF</i> decile	-3.531	[-0.86]	0.02	-3.668	[-0.98]	0.33
(16)	Asset weighted	-5.890	[-3.83]	0.21	-3.700	[-2.70]	0.38
(17)	Debt Issue weighted	-4.970	[-3.44]	0.16	-3.597	[-3.20]	0.39
(18)	Nonfinancials only	-15.178	[-4.02]	0.21	-13.234	[-3.64]	0.42
Alternate measures of credit quality							
(19)	Shumway distress	-18.751	[-3.60]	0.18	-15.560	[-3.31]	0.40
(20)	Interest coverage (Low – High)	-15.318	[-3.41]	0.16	-12.077	[-2.51]	0.39
(21)	Leverage (debt/assets)	-7.804	[-2.42]	0.10	-4.711	[-1.30]	0.34
(22)	CAPM σ	-8.696	[-3.25]	0.11	-4.310	[-0.91]	0.33
(23)	Size (small – big)	-10.967	[-1.71]	0.05	-3.222	[-0.50]	0.32
(24)	Age (young – old)	-15.141	[-2.38]	0.07	-16.263	[-2.18]	0.37
(25)	Dividends (nonpayer – payer)	-83.443	[-2.84]	0.12	-69.163	[-2.15]	0.37
Link to equity market							
(26)	Contemporaneous equity returns (<i>MKTRF</i>)	-13.409	[-5.32]	0.64	-12.232	[-3.80]	0.72
(27)	Contemporaneous Fama-French factors	-11.011	[-4.88]	0.69	-12.603	[-4.79]	0.76

Robustness checks on the time-series forecasting regression from Tables 2 and 3

$$r_{t+2}^{HY} = a + b \cdot ISS_t^{EDF} + c \cdot (y_{Lt}^G - y_{St}^G) + d \cdot y_{St}^G + e \cdot (y_{Lt}^{BBB} - y_{Lt}^G) + f \cdot r_{t+1}^{HY} + u_{t+2}.$$

The dependent variable is the two-year cumulative excess log return on high-yield bonds. The table reports the coefficient and *t*-statistic on the main variable of interest, ISS^{EDF} , as well as the regression R^2 . The multivariate specifications include controls for the term spread, short-rate, credit spread, and lagged excess high-yield returns. We adjustment the baseline regression in row (1) in a number of ways. Rows (2)–(5) present subsample results. In rows (6)–(11), we consider additional controls, including the output gap (Hodrick-Prescott filtered real GDP) in (6), other macroeconomic controls (industrial production, consumption growth, and a recession dummy) in (7), leads and lags of those macrocontrols in (8); the eight macrofactors from Ludvigson and Ng (2010) in (9), *cay* from Lettau and Ludvigson (2001) in (10); and the Cochrane-Piazzesi (2005) tent factor in (11). Rows (12)–(17) consider alternate constructions of ISS^{EDF} . Rows (18)–(24) consider alternate quality-related firm characteristics, including the Shumway (2001) bankruptcy hazard, interest coverage, leverage, idiosyncratic volatility, market capitalization, age, and payout policy. Finally, rows (25) and (26) control for contemporaneous equity market returns in (25) and for contemporaneous realizations of the Fama and French (1993) factors in (26). *t*-statistics (in brackets) are based on Newey-West (1997) standard errors allowing for serial correlation up to two lags.

6.9 Table 7: Predicting spread changes and defaults

Read: Table 7 uses issuer quality to forecast future BBB spread changes and future high-yield default rates.

- Poor issuer quality forecasts positive future spread changes.
- Poor issuer quality forecasts higher future defaults.

Why it matters. The low returns after credit booms come from both repricing and realized credit losses.

6.10 Table 8: Intermediary balance sheets

Read: Table 8 studies insurers, broker-dealers, and banks.

- Some intermediary-balance-sheet variables are correlated with issuer quality in the expected direction.
- But controlling for them hardly changes the predictive power of ISS^{EDF} .

Why it matters. The paper does not deny that intermediary capital matters. It shows that issuer quality contains information beyond these simple intermediary proxies.

Table 7
Forecasting changes in credit spreads and defaults

	Change in credit spreads						Defaults in year				
	1-year change		2-year change		3-year change		$t+1$	$t+2$	$t+3$		
	$\Delta_1(y_{t,t+1}^{BBB} - y_{t,t+1}^G)$		$\Delta_2(y_{t,t+2}^{BBB} - y_{t,t+2}^G)$		$\Delta_3(y_{t,t+3}^{BBB} - y_{t,t+3}^G)$						
ISS^{EDF}	0.609 (4.07)	0.352 (2.32)	0.925 (15.31)	0.449 (5.33)	0.419 (1.84)	0.879 (2.14)	1.502 (1.81)	1.352 (1.05)	2.990 (2.89)	3.322 (3.51)	2.488 (4.62)
$y_{t,t+1}^G - y_{t,t+1}^L$	-0.244 (-1.39)	-0.280 (-1.75)	-0.582 (-2.59)	-0.631 (-2.23)	-0.562 (-2.36)	-0.427 (-1.91)	0.281 (0.41)	0.261 (0.41)	-0.827 (-1.20)	-0.827 (-1.25)	-1.063 (-1.25)
$y_{t,t+1}^L$	0.024 (0.43)	0.043 (0.86)	-0.070 (-1.62)	0.009 (0.09)	-0.156 (-3.46)	-0.066 (-0.98)	0.005 (0.03)	0.005 (0.03)	-0.084 (-0.34)	-0.084 (-0.32)	-0.084 (-0.32)
$y_{t,t+1}^{BBB} - y_{t,t+1}^L$	-0.523 (-3.32)	-0.668 (-3.32)	-0.668 (-3.32)	-0.668 (-3.32)	-0.668 (-3.32)	-0.668 (-3.32)	0.824 (1.39)	0.824 (1.39)	1.401 (1.23)	1.401 (1.23)	1.501 (1.53)
r_{t+1}^{HY}	0.010 (1.53)	0.010 (1.53)	0.025 (1.94)	0.025 (1.94)	0.020 (1.30)	0.020 (1.30)	-0.084 (-2.09)	-0.084 (-2.09)	0.048 (1.12)	0.048 (1.12)	0.128 (2.01)
DEP^{HY}	0.052 (1.50)	0.052 (1.50)	0.093 (2.17)	0.093 (2.17)	-0.013 (-0.27)	0.786 (1.08)	0.618 (2.31)	0.614 (2.31)	0.681 (3.53)	0.359 (2.90)	0.538 (2.61)
R^2	0.11	0.17	0.45	0.22	0.36	0.60	0.18	0.36	0.55	0.37	0.56

Annual time-series forecasting regressions of changes in BBB-rated credit spreads:

$$\Delta_k(y_{t,t+k}^{BBB} - y_{t,t+k}^G) = a + b \cdot ISS_t^{EDF} + c \cdot (y_{t,t+k}^L - y_{t,t+k}^G) + d \cdot y_{t,t+k}^G + e \cdot (y_{t,t+k}^{BBB} - y_{t,t+k}^L) + f \cdot r_{t+k}^{HY} + g \cdot DEP_t^{HY} + u_{t+k}.$$

$$DEP_t^{HY} = a + b_1 \cdot ISS_t^{EDF} + b_2 \cdot DEP_t^{HY} + c \cdot (y_{t,t}^L - y_{t,t}^G) + d \cdot y_{t,t}^G + e \cdot (y_{t,t}^{BBB} - y_{t,t}^L) + f \cdot r_{t,t}^{HY} + u_{t,t}.$$

where Δ_k denotes the *k*-year difference; DEP_t^{HY} denotes the issuer-weighted default rate for high-yield bonds in year *t* + *k* from the annual Moody's default survey; $y^{BBB} - y^G$ is the credit spread; and ISS^{EDF} is the difference between the expected default frequency decile of high and low net debt issuers in each year. Control variables include the term spread, short-rate, credit spread, lagged excess high-yield returns, and the time-*t* default rate. *t*-statistics for *k*-period forecasting regressions are based on Newey-West (1997) standard errors allowing for serial correlation up to *k*-lags.

Table 8
Intermediary balance sheets and issuer quality

		Depvar: ISS^{EDF}		Depvar: r_{t+2}^{HY}		
		(1)	(2)	(3)	(4)	(5)
Insurer balance sheets	ISS^{EDF}			-13.102 [-3.41]		-14.251 [-3.69]
	$E/A_{Insurer}$	0.156 [3.58]	0.133 [2.54]	-4.104 [-3.78]	-2.058 [-1.68]	-1.336 [-0.76]
	$dA/A_{Insurer}$	-0.008 [-0.35]	-0.005 [-0.21]	-0.382 [-0.89]	-0.493 [-1.06]	0.382 [0.70]
	Controls	No	Yes	No	No	Yes
	R^2	0.21	0.48	0.15	0.30	0.33
Broker-dealer balance sheets	ISS^{EDF}			-16.006 [-5.25]		-14.579 [-4.20]
	E/A_{BD}	-0.017 [-0.98]	-0.018 [0.86]	0.160 [0.31]	-0.105 [-0.27]	-0.025 [-0.04]
	dA/A_{BD}	-0.005 [-1.60]	-0.003 [-1.06]	-0.008 [-0.08]	-0.088 [-0.96]	-0.035 [-0.49]
	Controls	No	Yes	No	No	Yes
	R^2	0.06	0.42	0.00	0.28	0.32
Bank balance sheets	ISS^{EDF}			-14.032 [-4.28]		-13.648 [-4.81]
	E/A_{Bank}	-0.013 [-0.40]	-0.032 [-0.96]	-0.558 [0.50]	-0.734 [0.88]	-0.208 [-0.15]
	dA/A_{Bank}	0.036 [1.48]	0.019 [0.71]	-1.285 [-2.64]	-0.780 [-1.60]	-0.768 [-1.58]
	Controls	No	Yes	No	No	Yes
	R^2	0.07	0.43	0.12	0.32	0.35
Lagged bank stock returns	ISS^{EDF}			-13.927 [-4.19]		-13.165 [-3.73]
	$R_{Bank,t-3,t}$	0.005 [2.54]	0.004 [3.36]	-0.105 [-3.60]	-0.039 [-1.18]	-0.068 [-2.40]
	Controls	No	Yes	No	No	Yes
	R^2	0.16	0.50	0.09	0.27	0.35
	R^2	0.14	0.41	0.10	0.28	0.32
Bank loan loss provisions	ISS^{EDF}			-13.594 [-4.30]		-13.913 [-4.42]
	$LoanLosses$	-0.397 [-1.72]	-0.139 [-0.59]	10.182 [2.00]	4.783 [1.34]	1.777 [0.36]
	Controls	No	Yes	No	No	Yes
	R^2	0.14	0.41	0.10	0.28	0.32
	R^2	0.14	0.41	0.10	0.28	0.32

This table explores the relationship between ISS^{EDF} and measures of intermediary balance sheet strength Z . The first two columns in each panel report the coefficient on Z_t from regressions of the form

$$ISS_t^{EDF} = a + b \cdot Z_t + c \cdot (y_{Lt}^G - y_{St}^G) + d \cdot y_{St}^G + e \cdot (y_{L,t+1}^{BBB} - y_{L,t}^G) + f \cdot r_{t+2}^{HY} + u_{t+k},$$

Columns (3) to (6) report the coefficients on ISS_t^{EDF} and Z_t from return forecasting regressions of the form

$$r_{t+2}^{HY} = a + b \cdot ISS_t^{EDF} + b \cdot Z_t + c \cdot (y_{Lt}^G - y_{St}^G) + d \cdot y_{St}^G + e \cdot (y_{L,t+1}^{BBB} - y_{L,t}^G) + f \cdot r_{t+2}^{HY} + u_{t+2},$$

where r_{t+2}^{HY} denotes the two-year excess return on high-yield bonds. The regressions alternately omit ISS^{EDF} and include the term spread, short-rate, credit spread, and lagged excess high-yield returns as controls. Measures of intermediary balance sheet strength, summarized in Table 1, include insurer equity capital E/A (total assets minus liabilities all over assets) and balance sheet growth dA/A (year-over-year percentage change in total assets), broker-dealer equity capital E/A and balance sheet growth dA/A , bank equity capital E/A and balance sheet growth dA/A , lagged equal weighted returns on bank stocks based on Fama and French 48-industry classifications, and bank loan loss provisions scaled by total loans and leases. t -statistics for two-year forecasting regressions (in brackets) are based on Newey-West (1987) standard errors allowing for serial correlation up to two lags.

Table 9
Determinants of issuer quality

		ISS^{EDF}					$\Delta_1 ISS^{EDF}$					$\Delta_2 ISS^{EDF}$				
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Levels	y_{Lt}^G	-0.047 [-1.63]		-0.036 [-1.22]	-0.035 [-1.14]	-0.026 [-0.99]										
	$(y_{Lt}^G - y_{St}^G)$	-0.277 [-4.91]		-0.233 [-3.41]	-0.218 [-2.98]	-0.204 [-2.61]										
	r_{t+1}^{HY}	0.014 (1.57)	0.015 (2.78)	0.009 (1.42)	0.016 (3.14)											
	DEP_t	-0.076 [-3.11]	-0.013 [-0.59]	-0.017 [-0.68]	-0.010 [-0.40]											
	$\Delta_1 (y_{Lt}^G - y_{St}^G)$						-0.107 [-2.41]	-0.072 [-1.65]	-0.077 [-2.04]	-0.078 [-2.58]						
1-year changes	$\Delta_1 (y_{Lt}^G - y_{St}^G)$						-0.335 [-3.99]	-0.173 [-2.74]	-0.165 [-2.71]	-0.137 [-2.05]						
	r_{t+1}^{HY}						0.031 (6.60)	0.025 (5.90)	0.014 (2.24)	0.018 (3.25)						
	$\Delta_1 DEP_t$						-0.043 [-1.74]	-0.025 [-1.11]	-0.031 [-1.36]	-0.020 [-1.11]						
	$\Delta_2 (y_{Lt}^G - y_{St}^G)$										-0.134 [-3.31]	-0.062 [-1.94]	-0.061 [-1.70]	-0.084 [-1.92]		
	$r_{t+3,t+1}^{HY}$										-0.410 [-7.63]	-0.164 [-2.52]	-0.154 [-2.11]	-0.168 [-1.53]		
2-year changes	$\Delta_2 (y_{Lt}^G - y_{St}^G)$										0.036 (6.91)	0.027 (4.47)	0.019 (2.77)	0.018 (2.65)		
	$\Delta_2 DEP_t$										-0.065 [-3.77]	-0.069 [-2.72]	-0.064 [-2.78]	-0.083 [-2.45]		
	Other controls	None	None	None	Lagged MKT	Macro Vars	None	None	None	Lagged MKT	Macro Vars	None	None	None	Lagged MKT	Macro Vars
	R^2	0.44	0.34	0.53	0.55	0.61	0.36	0.49	0.55	0.62	0.74	0.47	0.59	0.62	0.64	0.75

Time-series regressions of issuer quality ISS^{EDF} on levels and past changes of interest rates:

$$ISS_t^{EDF} = a + b \cdot y_{Lt}^G + c \cdot (y_{Lt}^G - y_{St}^G) + d \cdot r_{t+1}^{HY} + e \cdot DEP_t + u_t, \text{ or } \Delta_1 ISS_t^{EDF} = a + b \cdot \Delta_1 y_{Lt}^G + c \cdot \Delta_1 (y_{Lt}^G - y_{St}^G) + d \cdot r_{t+1,t-1}^{HY} + e \cdot \Delta_1 DEP_t + \Delta_1 u_t.$$

y_{Lt}^G denotes the short-term Treasury-bill yield; $y_{St}^G - y_{Lt}^G$ denotes the term spread; DEP_t is the issuer-weighted high-yield default rate from Moody's; $r_{t+1,t-1}^{HY}$ is the excess high-yield return; and Δ_k denotes the k -year difference. In Columns (1) to (5) we regress the level of ISS^{EDF} on a number of covariates; Columns (6) to (10) repeat this analysis in first differences; and Columns (11) to (15) repeat this analysis in second differences. In the last two columns in each block, we add additional controls for lagged stock market returns and macroeconomic variables (the growth in industrial product, real consumption growth, and a recession indicator). Robust t -statistics are shown in brackets.

6.11 Table 9: Determinants of issuer quality

Read: Table 9 regresses issuer quality on interest rates, the term spread, past high-yield returns, and past high-yield default rates.

- Low levels or declines in short rates and the term spread are associated with poorer future issuer quality.
- Low past defaults and high past credit returns also predict poorer issuer quality.
- These relations are particularly strong in changes.

Why it matters. The table is the main evidence for the paper’s “reaching for yield” and overextrapolation interpretations.

7 Short summary

- **Main object.** Measure the quality of debt issuers, not just the quantity of debt issued.
- **Main empirical result.** Poor issuer quality forecasts low future excess corporate bond returns, especially for high-yield bonds.
- **Main conceptual result.** Because low-quality firms are more sensitive to changes in the price of credit risk, the composition of debt issuance is a better signal of expected returns than aggregate issuance.
- **Main robustness result.** The predictive relation survives standard bond-market controls, macro controls, and alternative issuer-quality constructions.
- **Main decomposition.** Poor issuer quality predicts both spread widening and future defaults.
- **Main interpretation.** Rational discount-rate variation may explain part of the pattern, but the results also point toward intermediary frictions, reaching for yield, and possibly investor overextrapolation.
- **Main policy lesson.** Credit overheating should be monitored through *who* is borrowing, not just *how much* is being borrowed.

8 Conclusion

8.1 Core conclusion

Greenwood and Hanson show that the quality mix of corporate debt issuance is a powerful state variable for credit markets. When weak firms account for an unusually large share of debt issuance, future corporate bond returns are low.

8.2 Economic intuition

If broad financing conditions improve because investors demand less compensation for bearing credit risk, low-quality firms benefit the most. They issue more aggressively, so a deterioration in issuer

quality reveals unusually cheap credit. That cheap credit is later followed by poor bondholder returns.

8.3 Takeaway

The paper changes how to think about credit booms. The most informative sign of overheating is not just rapid credit growth or narrow spreads. It is the combination of *easy financing and poor borrower quality*. In that sense, issuer quality is a compact summary statistic for the state of the credit cycle.